

Resonance Bastion

Tablet target and optimization • 0.2.9 • 2026-10-08

Your screenshots resolve the device/runtime question. Development continued with a measured rendering improvement; game scope, artwork detail, progression and continuous 360° movement remain intact. This supplements the preceding four-page development review, also included in the ZIP.

Confirmed from screenshots	Value
Model / Android	Lenovo TB-X606F / Android 10
RAM / storage	4.00 GB / 64.00 GB
CPU / physical panel	MediaTek Helio P22T / 1920×1200
Installed browser reported	Chrome 154.0.8037.126

The browser exceeds researched Chrome version minima for BigInt, crypto.randomUUID, Object.hasOwn and structuredClone (67/92/93/98). UUID still requires a secure context. This establishes API version compatibility, not actual tablet storage, audio or GPU reliability. Desktop tests use Chromium 151; panel resolution alone does not establish browser CSS viewport or pixel ratio.

One measured bottleneck corrected

Camera, pinch, placement and resize redraws previously destroyed and recreated all terrain images, object images and text labels. They now reuse scene objects up to peak visible demand. Unused objects become hidden; active picking stays correct. Changed roles reset frame, anchor, tint, opacity, scale and depth. Scene disposal releases the retained objects.

81 redraws / case	New images before → after	Median draw ms before → after
Portrait Base	261,135 → 1	12.7 → 2.1
Portrait Field	292,734 → 0	15.4 → 2.0
Portrait Close	65,286 → 0	2.4 → 0.7
Portrait Ghost	261,216 → 1	13.0 → 1.4
Landscape Field	292,653 → 0	16.8 → 1.6
Landscape Close/Ghost	59,049 → 0	2.9 → 0.7

These are actual shared-renderer desktop draw-submission measurements, excluding screenshot work. They are not GPU frame times or Lenovo frame-rate results. Peak pool memory still needs physical measurement. No simulation or sprite-resolution reduction was used.

Verification and playable tablet review

The rendering is preserved exactly

All six captured before/after PNGs match byte-for-byte, also checked independently as RGB pixels. Camera and selection bounds match. The optimized cases destroy zero objects during the 81 redraws; only two cases need one additional image as the camera discovers slightly greater demand.

Repeated native-browser pan/zoom/facing/ghost cycles produce no second-cycle allocations or child-count growth. Surplus sprites stay hidden and excluded from picking. Ghost/role/text-style transitions restore identical pixels. Scene disposal removes the canvas and clears retained public collections.

Full current-slice regression passes

All 134 unit tests in 30 files and all 49 browser checks pass without retries or skips. Typecheck, lint, formatting, deterministic scenario and production build pass. The full suite covers real app touch/pinch/cancel, four views, scaling/reflow, renderer recovery, practice, paid results, backup/import and exactly-once commits. Retained v1/v2/v3 replay hashes remain exact.

The evidence ZIP retains target/reference records, before/after allocation and timing observations, camera/picking state, screenshots and source/log hashes. The original device screenshots and unrelated personal/network details are not redistributed.

Physical-device validation needs a playable origin

A stable HTTPS game URL is needed to measure actual tablet GPU/frame behavior, touch latency, audio/background lifecycle, save reliability and backup download/reimport. The reviewed PDF and ZIP can be opened now, but they are not a hosted playable game. No physical performance claim is made.

A concrete Free-compatible GitHub Pages preview is prepared

The manual-only workflow pins the tested 0.2.9 source, uses official action commit pins and a standard free public-repository Ubuntu runner, retains its small static artifact for one day and has no automatic push trigger. It is prepared for approval, not installed or dispatched. Main and repository hosting remain unchanged.

Publishing this development checkpoint as a public playable preview requires your approval. After publication and real HTTPS/MIME/refresh checks, the tablet pass can test 1×/2×/4×, orientation and all travel directions, pause/audio, background-return without catch-up, checkpoint restart and Android backup/reimport. Any missing Pages settings permission will be identified precisely; no paid service is needed.

Independent development continues

The next individual Engineer contract separates purchased and siege-temporary trap charges, enforces six uninterrupted seconds and eight completed additions per agent, and cancels interrupted work without free permanent stock. Full mobile repair/action priorities, state-machine integration and natural per-asset animation still require their own evidence. The wider roster/campaign and full balance remain open.

Engineer rearm: the next individual contract

Development continued beyond the rendering checkpoint. An isolated Engineer finite-stock and uninterrupted-rearm kernel is implemented and tested. The Engineer is not yet enabled in the game; mobile priorities, repair/fire exclusion, captured profiles and original animation still require their separate increments.

Exact stock and timing rules

Two visual engineers belong to one owned agent: one rearm action and eight completed additions per siege, not sixteen. One trap can reserve one Engineer. A channel begun at tick T completes at T+360 (six full seconds); there is no immediate charge or level-based time/count improvement. Every authoritative tick is observed, irrespective of game speed.

Temporary combat charges are spent before purchased charges. Total stock never exceeds its initial cap. Results can retain only the actual remaining purchased stock; unused temporary charges cannot become permanent stock or a refund. This kernel makes no campaign, currency, body or XP writes.

Interruptions are scrutinized at the completion boundary

Movement, invalid range/LOS, source or trap death, incoming damage, unsafe nearby threats, being under attack or a higher-priority action cancels the unfinished rearm. Missing observations and filled traps release their reservation. Cancellation spends no completed addition; restarting needs a new full six seconds. Completed rearms survive later interruptions.

The entire completion batch is validated before any channel/count update. Invalid observations cannot complete one Engineer while failing another. Retrying the corrected tick completes once; repeated or skipped committed ticks are rejected. The live integration must supply real spatial/perception/death observations and apply returned temporary stock in the correct fixed-step phase.

Evidence

Nine new unit cases cover all initial caps 1-8 and starting purchased-stock counts, temporary-first consumption, tick 359/360, all interruption flags at completion, restart, reservation conflicts, exhausted allowance and invalid-batch retry. The complete suite passes 143 unit tests in 31 files, plus typecheck, lint, formatting, scenario and build.

All 38 production files are byte-for-byte identical to the preceding playable build, which passed 49 browser checks. No broad browser rerun was needed for an unimported kernel. Retained v1/v2/v3 replay regressions still pass. The source and output hashes are included in this ZIP.

Remaining work and publication input

Next integrate the actual mobile repair/action planner under a new simulation version, then prove its movement, injury-scaled output, shared channels, stock/result accounting and natural pose/contact transitions. Full blueprint scope remains intact. Separately, approval to enable the prepared public GitHub Pages tablet preview is pending; physical tablet performance remains unmeasured.